

Mock BE Practical Viva - Detailed Notes

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1) Feature Transformation - Principal Component Analysis (PCA)

Feature Transformation means converting data from its original feature space into a new one. PCA (Principal Component Analysis) is a dimensionality reduction technique that transforms correlated variables into uncorrelated ones (Principal Components). It helps reduce complexity while keeping most of the data variance. PCA finds directions (components) that capture the most variance in data.

Steps:

1. Standardize data (mean=0, variance=1)
2. Compute covariance matrix
3. Find eigenvalues and eigenvectors
4. Sort eigenvalues and select top components
5. Transform data into new feature space

Used for distinguishing red and white wine using top PCs.

Advantages: Removes redundancy, improves visualization, and speeds up computation.

2) Regression Analysis - Uber Fare Prediction

Regression predicts continuous output variables. Linear Regression models a linear relation between dependent and independent variables.

Ridge Regression (L2 regularization) penalizes large coefficients to prevent overfitting.

Lasso Regression (L1 regularization) can shrink coefficients to zero for feature selection.

Steps:

1. Pre-process data, handle missing values
2. Detect outliers (boxplots/z-score)
3. Check correlation between variables
4. Train-test split
5. Implement Linear, Ridge, and Lasso models
6. Evaluate using R², RMSE, and MAE

Goal: Compare models to predict Uber fare most accurately.

3) Classification Analysis - K-Nearest Neighbors (KNN)

KNN is a supervised classification algorithm based on similarity. It predicts a class based on majority vote among K nearest neighbors.

Steps:

1. Choose K (odd number preferred)
2. Compute distances (Euclidean/Manhattan)
3. Select nearest K neighbors
4. Assign most common class among them

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix.

Use Case: Predicting purchase behavior in social network ads.

4) Clustering Analysis - K-Means

K-Means is an unsupervised algorithm grouping similar data points into K clusters.

Steps:

1. Choose K (number of clusters)
2. Randomly assign centroids
3. Assign points to nearest centroid
4. Recalculate centroids
5. Repeat until stable

Elbow Method helps choose optimal K (using WCSS vs. K plot).

Use Case: Classifying Iris flowers into species.

5) Ensemble Learning - Random Forest, AdaBoost, GBM, XGBoost

Ensemble learning combines multiple models for better accuracy.

A) Random Forest: Many decision trees built on random samples (bagging). Final output = majority vote.

B) Boosting Algorithms: Build sequential models improving previous errors.

- AdaBoost: Adjusts sample weights after each iteration.
- GBM: Minimizes loss via gradient descent.
- XGBoost: Optimized GBM with regularization and faster execution.

Metrics: Accuracy, Precision, Recall, ROC-AUC.

Used for car safety prediction and Iris classification.

6) Reinforcement Learning (Maze, Taxi, or Tic-Tac-Toe)

Reinforcement Learning trains an agent to take actions in an environment to maximize rewards.

Key Elements: Agent, Environment, State, Action, Reward, Policy, Value Function.

Q-Learning Formula:

$$Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where α =learning rate, γ =discount factor.

Examples:

- Maze: Agent learns to reach goal.
- Taxi Problem: Agent picks and drops passengers.
- Tic-Tac-Toe: Agent learns optimal moves.

Goal: Learn by trial and error through rewards and penalties.

7) Data Cleaning and Preparation - Telecom Churn

Data cleaning ensures quality and consistency of data.

Steps:

1. Import dataset
2. Handle missing values (mean/median/mode)
3. Remove duplicates
4. Fix inconsistencies in format
5. Convert data types
6. Detect and handle outliers
7. Perform feature engineering (new useful variables)
8. Normalize or scale data
9. Split data into train/test sets
10. Export cleaned dataset

Goal: Identify key factors influencing churn (e.g., contract type, tenure).

8) Data Wrangling - Real Estate Market

Data wrangling transforms raw data for analysis.

Steps:

1. Import dataset, rename columns
2. Handle missing/outlier values
3. Merge datasets (e.g., demographics or amenities)
4. Filter data by property type/location

5. Encode categorical data (label/one-hot)

6. Aggregate statistics (avg price by area)

7. Handle outliers

Goal: Prepare clean dataset for predicting property prices.

9) Data Visualization - Air Quality Index (AQI)

Data visualization uses plots to reveal patterns and trends.

Library: Matplotlib

Plots:

- Line Plot: AQI trends over time
- Bar Chart: Pollutant comparison
- Box Plot: AQI distribution
- Scatter/Bubble Plot: Relation between AQI and pollutants

Goal: Analyze pollution patterns and identify major pollutants.

Customize plots with labels, titles, legends, and colors.

10) Data Aggregation - Retail Sales

Data aggregation groups and summarizes data for insights.

Steps:

1. Group by region, product, etc.
2. Compute total or average sales
3. Visualize (bar/pie charts)
4. Identify top regions or products
5. Create stacked bar plots for product-category analysis

Goal: Understand sales trends and regional performance.

Viva Tips:

- Always define, explain, and provide examples.
- Revise metrics: Accuracy, Precision, Recall, F1, RMSE, R2.
- Understand differences: Supervised, Unsupervised, and Reinforcement Learning.
- Highlight importance of data preprocessing and visualization.